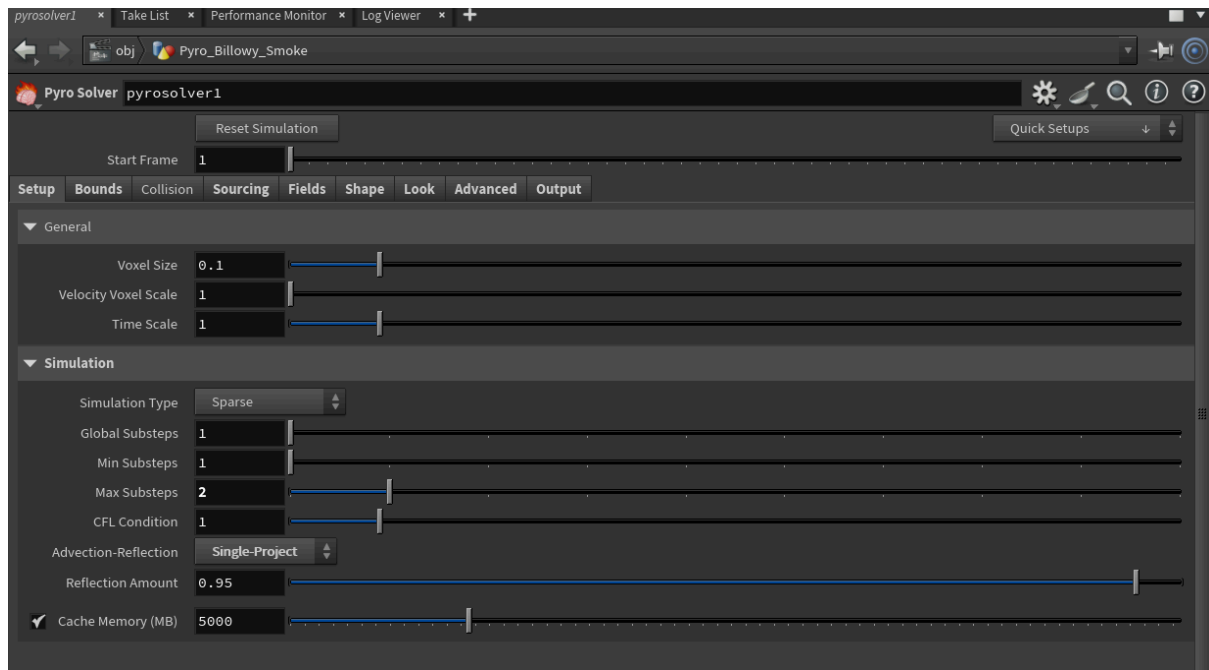


Simple Guide on using Venkys Turbulence Microsolver

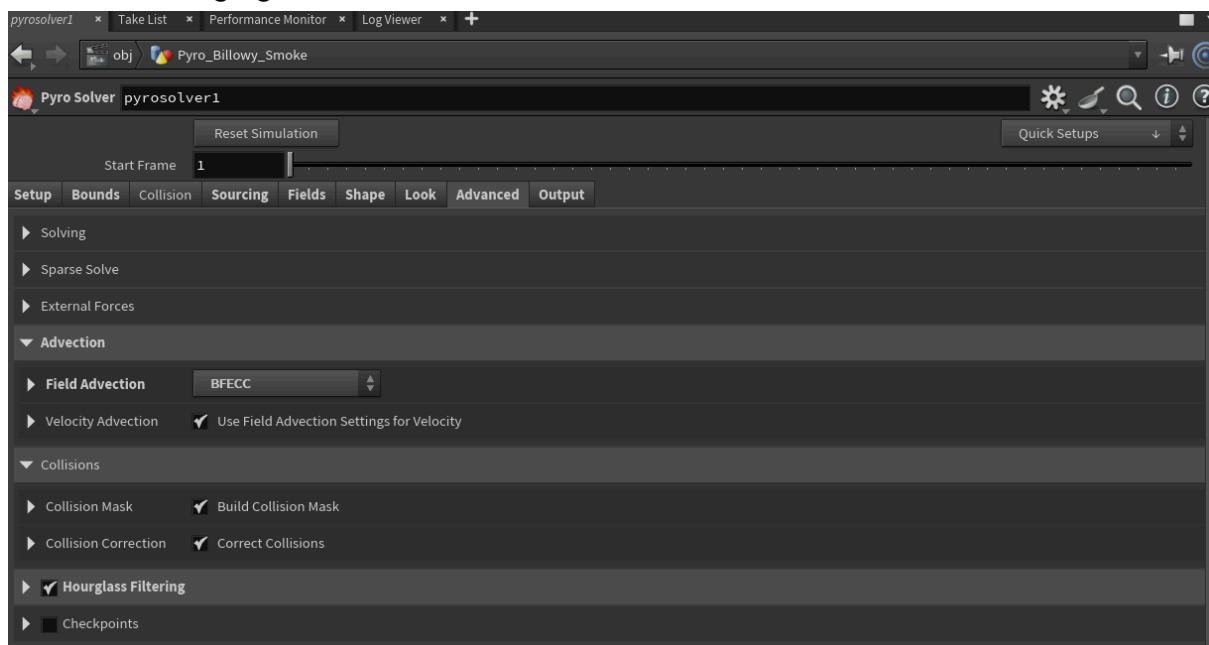
Here I am using the Pyro Billowy smoke shelf tool in H21.

When I first have the preset loaded, I do a couple of main settings that help overall precision of the pyro solver.

First is using 2 max substeps with Single Project in Advection-Reflection



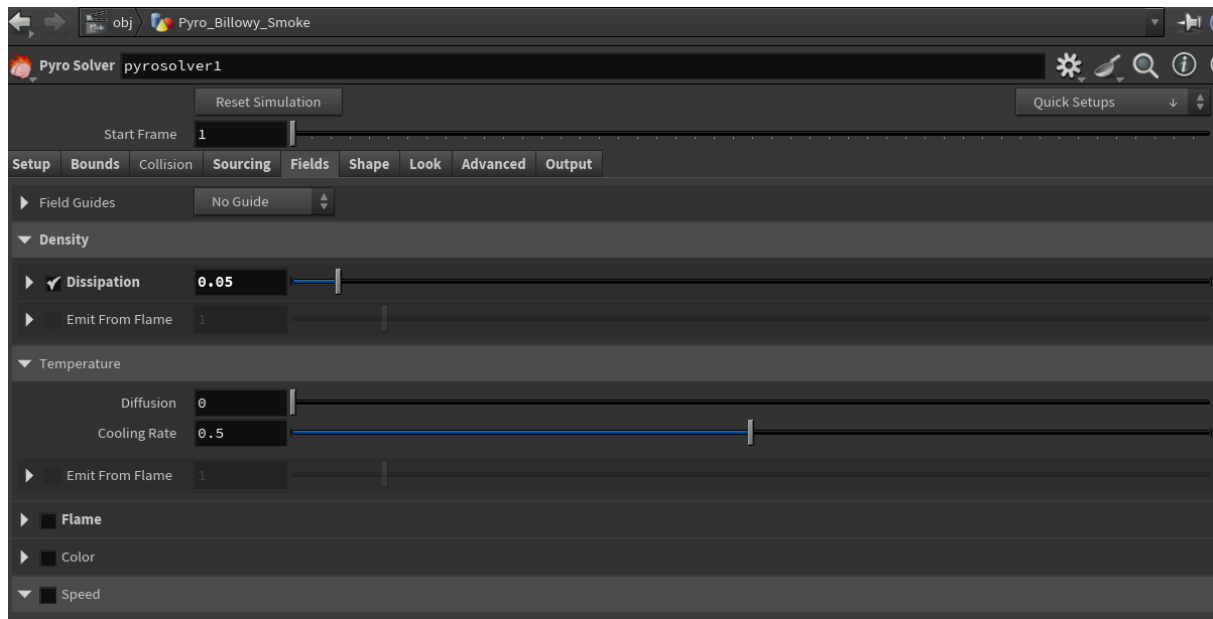
Second is changing the Advection Scheme to BFECC.



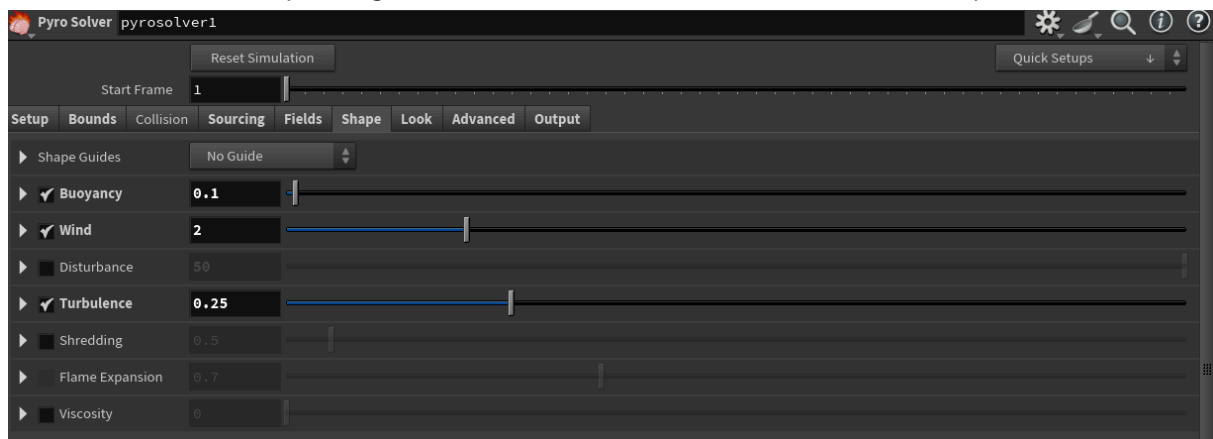
Also unchecked Hourglass Filtering.

And disabled all other output fields except density for faster viewport performance.

Now to optimize the solver I have unchecked, Flame and Speed in Fields section.

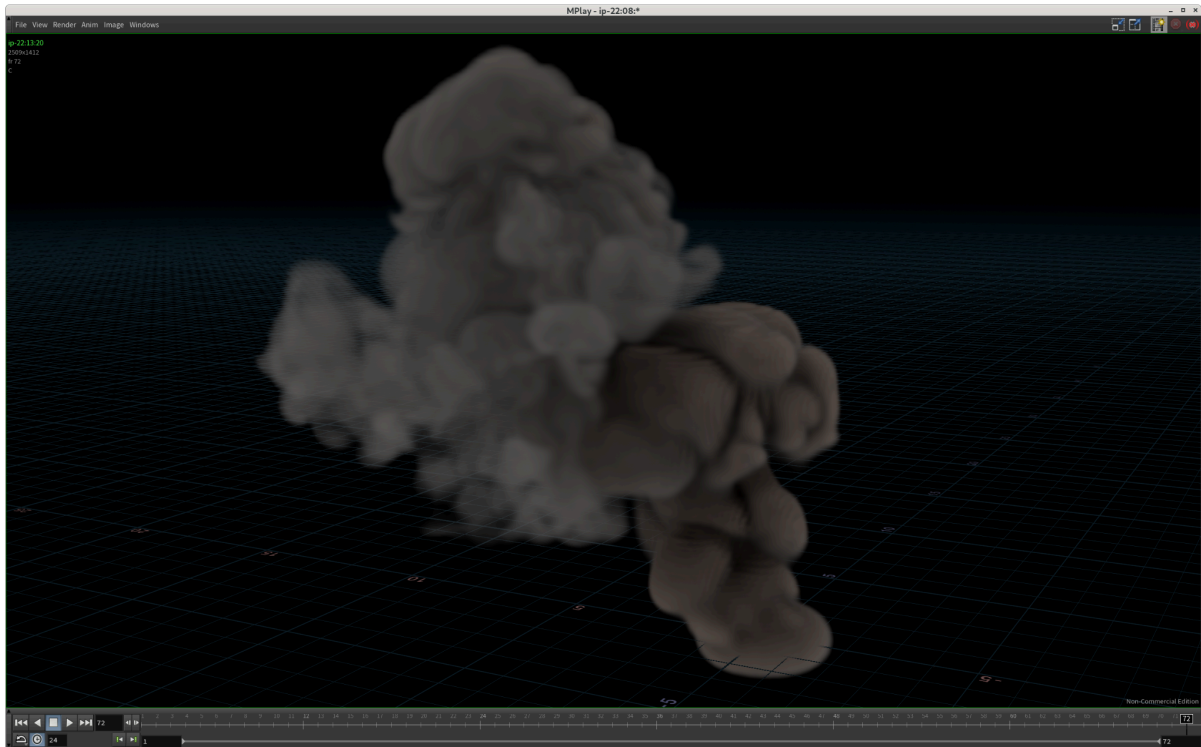


Since our solver is replacing the Disturbance, I unchecked it in the Shape section.

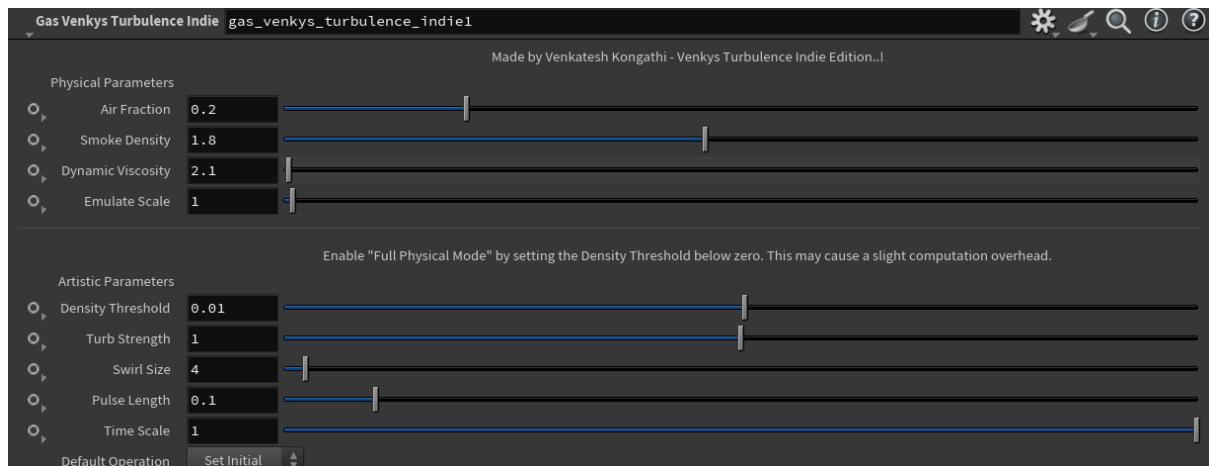


I have made a flipbook of 72 frames with current settings for reference, let's call it a vanilla flipbook.

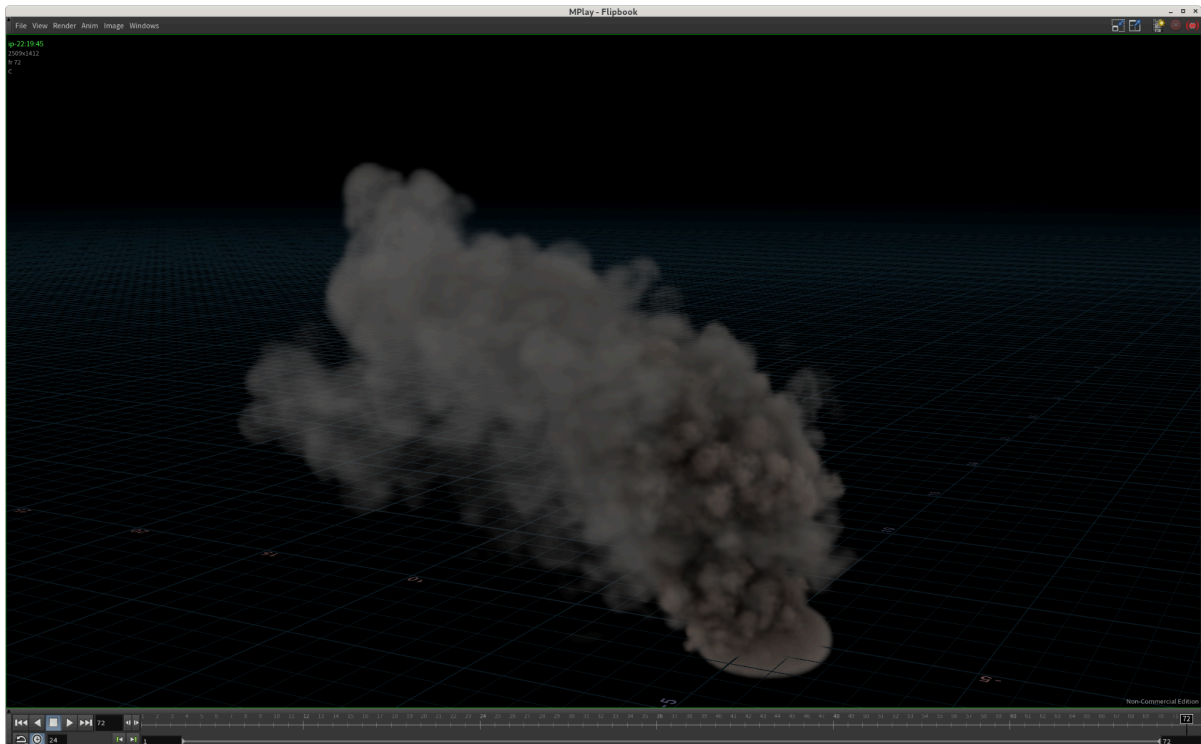
It gives a good idea on what physical parameters you have to tweak on Venky's Turbulence microsolver.



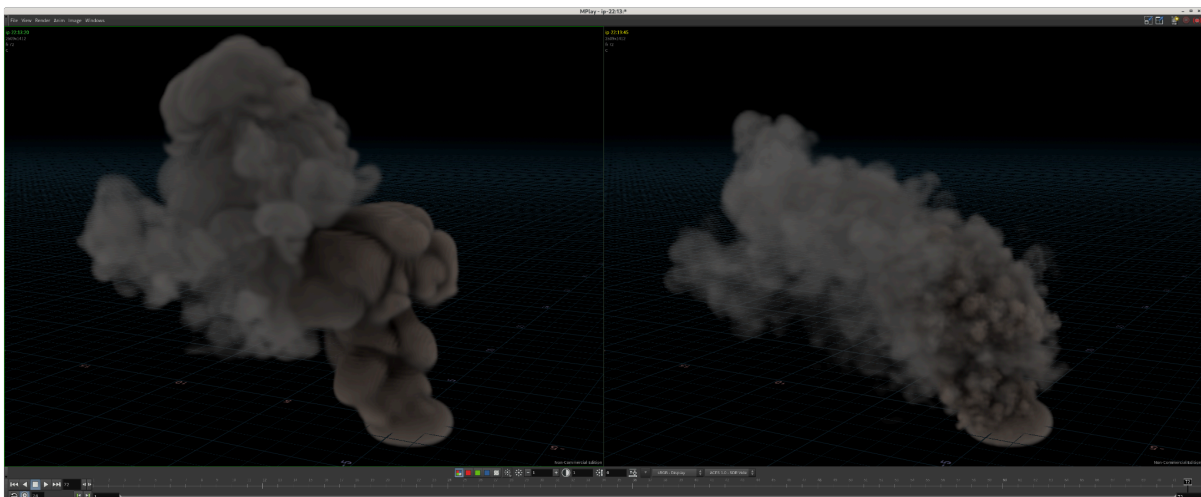
I dropped down Venkys Turbulence solver with its default settings



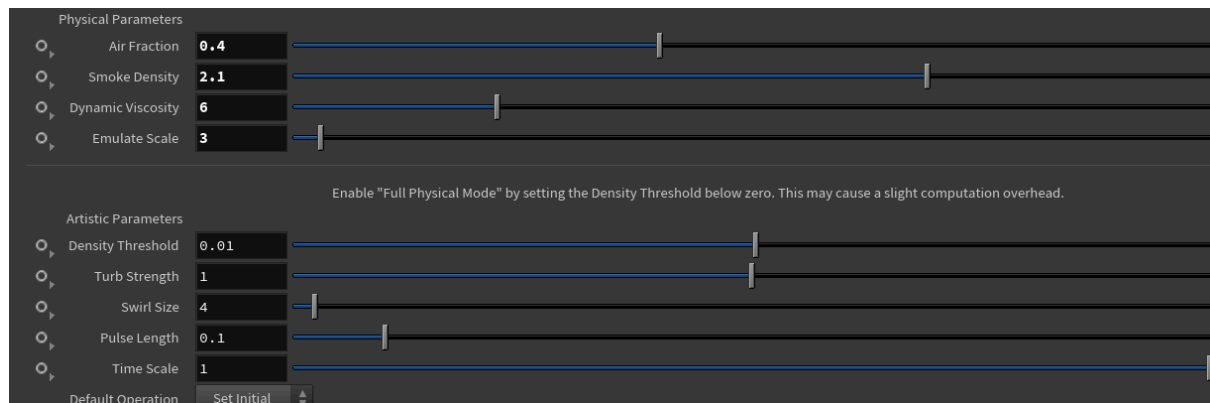
I quickly ran a new flipbook sequence to see its effect. It doesn't look that nice!



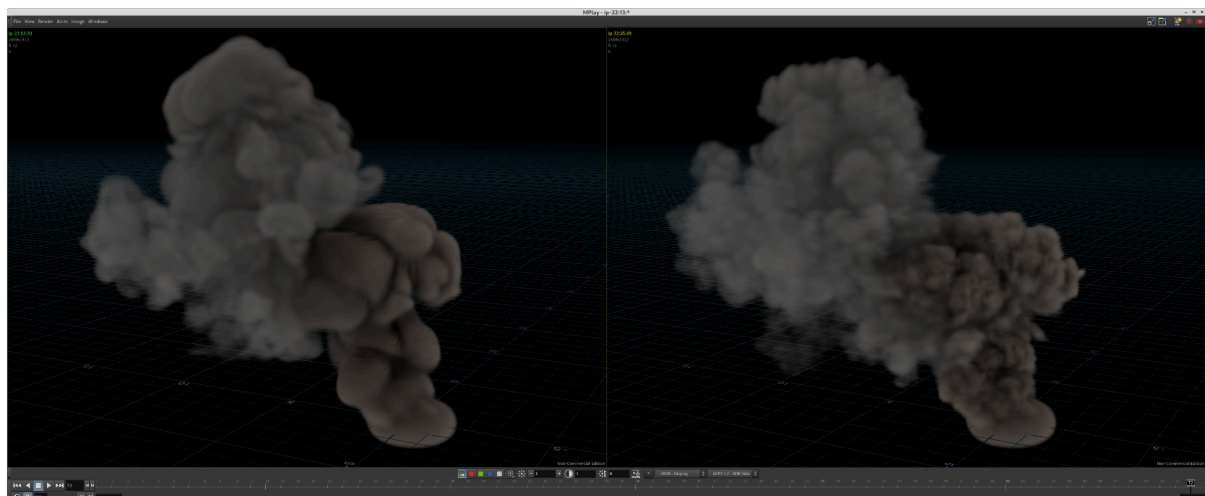
If we compare this with the vanilla flipbook it doesn't match the silhouette. Which means our dynamic viscosity value is not matching the scale of the smoke we are trying to simulate.



Since we are using a density threshold above zero, which means the turbulence is partially physical, I will raise the AirFactor to 0.4 for better breakup. Our smoke looks quite dense, so I raised the smoke density to 2.1 to match the look. Based on the vanilla flipbook, the smoke travels almost 10 meters and 6 meters high. Which means our Dynamic Viscosity needs to be higher and also we can use higher around 6 and emulate scale value around 3.

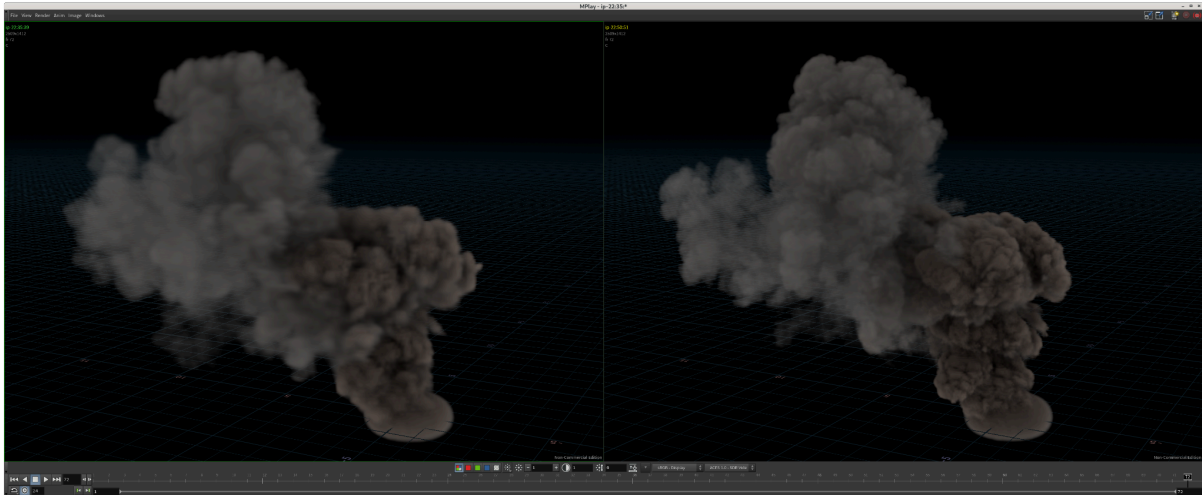


Now I ran a new flipbook sequence to compare with vanilla. As you can see the smoke almost matches the vanilla version in terms of silhouette while adding a ton of organic details. Since our turbulence adds natural air drag, it won't exactly grow or match in shape to the vanilla version.

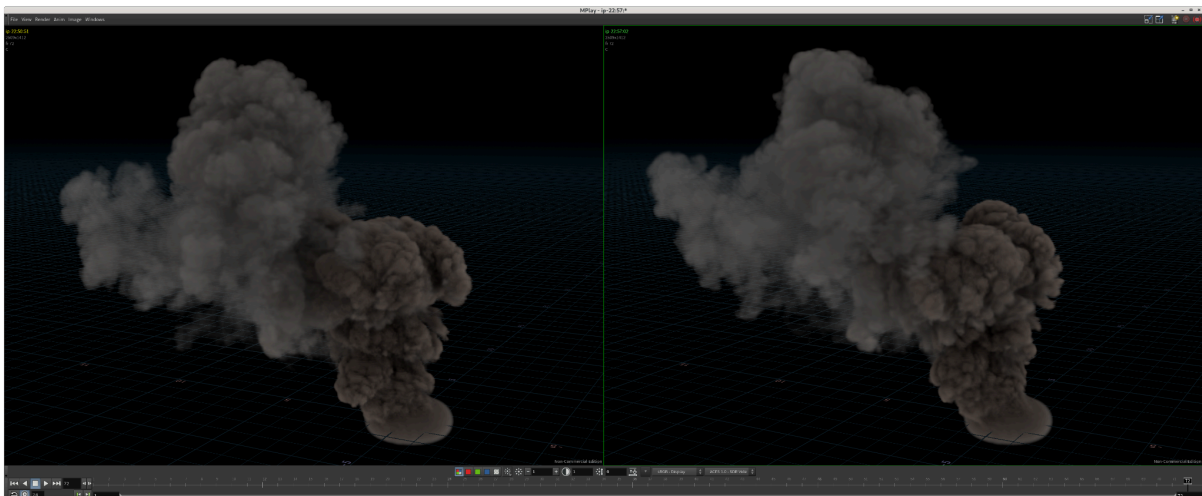


As you can see it's quite simple to use once you have the vanilla sim for reference.

I have changed voxel resolution to 0.05 from 0.1 to and did a new flipbook.
As you can see, it produces the maximum amount of details for that resolution while maintaining silhouette.



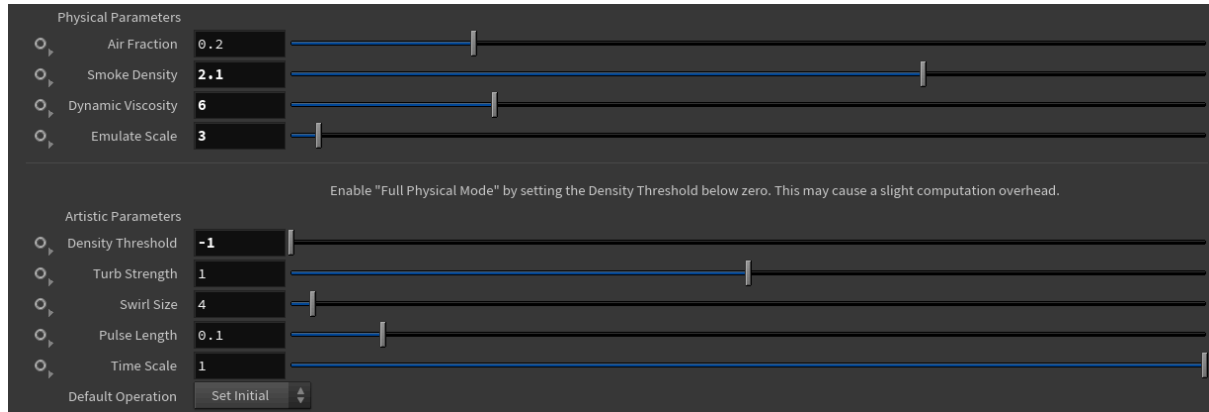
I also did a flipbook with 3 max substeps to show you guys the difference in details.
Pick it to your liking.



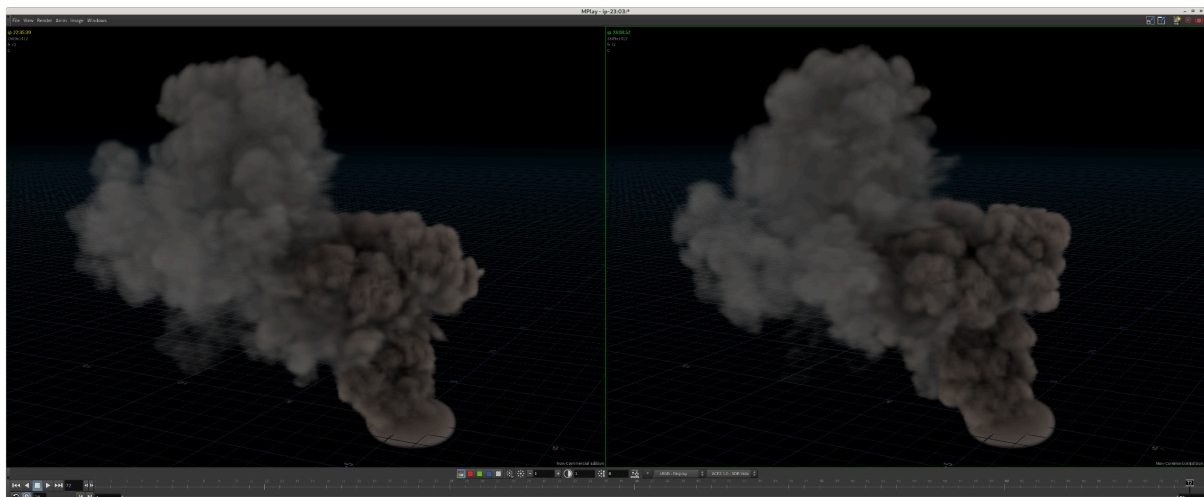
Now I will go back to the lower resolution sim with voxel size of 0.1 and change max substeps to 2.

Let's look into full physical mode.

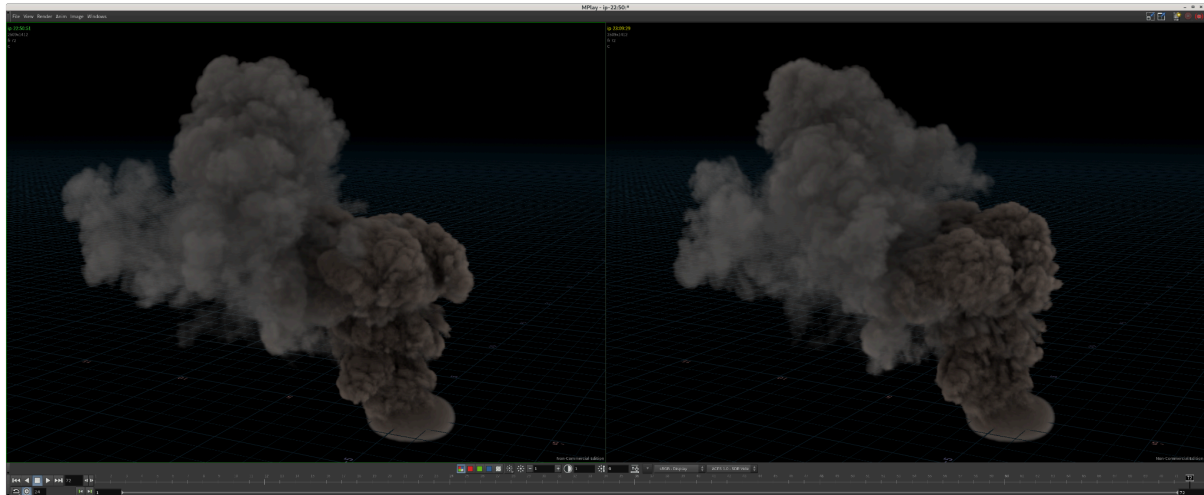
For that bring down the Air Factor to 0.2 and set density threshold to -1.



Full physical mode (right) has more surface detail and adds natural diffusion to the smoke as it raises that makes it more puffy. You can see that in this comparison with partial mode (left).



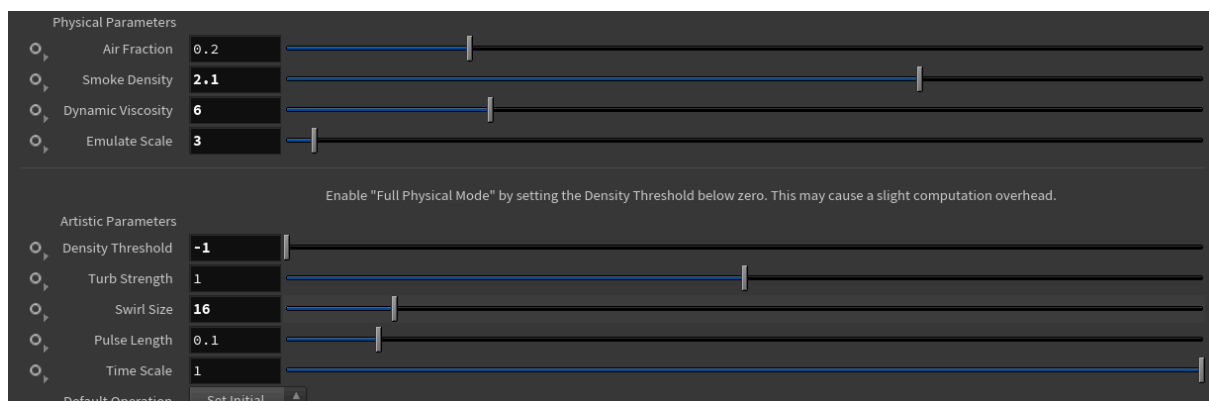
I also ran a high resolution version to compare between partial physical (left) and full physical (right). You can notice a lot of surface details.



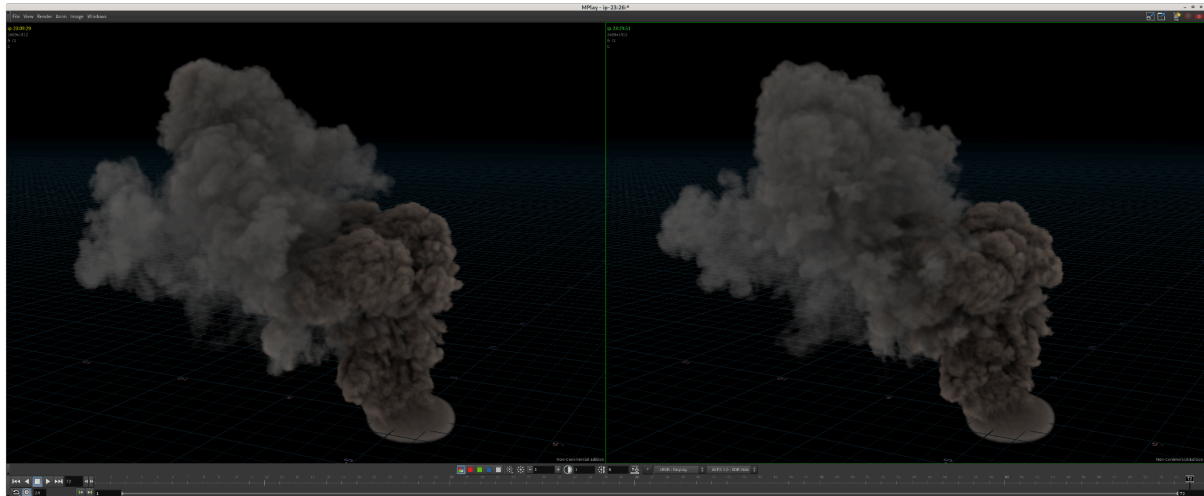
Now let's see what swirl size does.

Swirl Size 4 will give maximum details in a given voxel resolution with minimal artefacts. This helps maintain the silhouette with the vanilla simulation.

Lets change the Swirl Size to 16, which means each swirl occupies around 16 voxels across.



I ran a new flipbook sequence with the same high res and full physical settings. As you can see the silhouette and the details look a bit different. This way you can add some interesting breakup to your simulations if needed.



Key Points:

Turb Strength: as you can see I have never touched Turb Strength parameter at all. It's there if you want to address some notes like reduce turbulence by like 30% or raise it by 20%.

If you are doing an explosion or something that occupies a large area, higher values in dynamic viscosity works better. Something around 10-20.

If your explosion has a lot of expansion and it is not expanding after the adding turbulence it means your dynamic viscosity is too low for a given smoke density value.

Use Emulate scale to add more details when you have high dynamic viscosity.

Use pulse length to play around for different looks.

When the density threshold is positive and you see artefacts, you might want to raise it slightly to help remove those.

Thank you for reading through, I hope this helps.

Enjoy!